

```
1) #include <iostream>

#include <string>

using namespace std;

class Account {
private:
    int accountNumber;
    string customerName;
    double balance;
public:
    Account() {
        accountNumber = 0;
        customerName = "Unknown";
        balance = 0.0;
        cout << "Default constructor called." << endl;
    }
    Account(int accNo, string name, double bal) {
        accountNumber = accNo;
        customerName = name;
        balance = bal;
        cout << "Parameterized constructor called." << endl;
    }
    Account(const Account &obj) {
        accountNumber = obj.accountNumber;
        customerName = obj.customerName;
        balance = obj.balance;
        cout << "Copy constructor called." << endl;
    }
    void display() {
```

```

        cout << "Account Number : " << accountNumber << endl;
        cout << "Customer Name : " << customerName << endl;
        cout << "Balance      : Rs. " << balance << endl;
        cout << "-----" << endl;
    }
};

```

```

int main() {
    Account account1;
    account1.display();

    Account account2(1001, "Thanuja", 25000.50);
    account2.display();

    Account account3(account2);
    account3.display();

    return 0;
}

```

The screenshot shows a C++ program editor with a dark theme. The code defines an `Account` class with two constructors: a default constructor and a parameterized constructor. The `main` function creates three `Account` objects: `account1` (default), `account2` (parameterized), and `account3` (copy of `account2`). Each object's `display` method is called. The editor includes a 'PROGRAM EDITOR' header, a language dropdown set to 'C++', and 'Run' and 'Save' buttons. On the right, an 'INPUT' panel contains the text 'Your INPUT goes here!'.

```

7   int accountNumber;
8   string customerName;
9   double balance;
10
11 public:
12     Account() {
13         accountNumber = 0;
14         customerName = "Unknown";
15         balance = 0.0;
16         cout << "Default constructor called." << endl;
17     }
18     Account(int accNo, string name, double bal) {
19         accountNumber = accNo;
20         customerName = name;
21         balance = bal;
22         cout << "Parameterized constructor called." << endl;

```

```
Default constructor called.  
Account Number : 0  
Customer Name  : Unknown  
Balance        : Rs. 0  
-----  
Parameterized constructor called.  
Account Number : 1001  
Customer Name  : Thanuja  
Balance        : Rs. 25000.5
```

2)

```
#include <iostream>  
  
#include <string>  
  
using namespace std;  
  
class Product  
{  
private:  
    int productId;  
    string productName;  
    int stock;  
  
public:  
    Product(int id, string name, int quantity)  
    {  
        productId = id;  
        productName = name;  
        stock = quantity;  
        cout << "Parameterized Constructor called for "  
             << productName << endl;  
    }  
}
```

```

Product(const Product &p)
{
    productId = p.productId;
    productName = p.productName;
    stock = p.stock;

    cout << "Copy Constructor called for "
          << productName << endl;
}

Product operator+(const Product &p)
{
    Product temp(0, "Combined Product", stock + p.stock);
    return temp;
}

void display()
{
    cout << "\nProduct ID   : " << productId << endl;
    cout << "Product Name : " << productName << endl;
    cout << "Stock       : " << stock << endl;
}

~Product()
{
    cout << "Destructor called for "
          << productName << endl;
}

};

int main()
{
    cout << "==== INVENTORY PRODUCT MANAGEMENT SYSTEM =====\n";

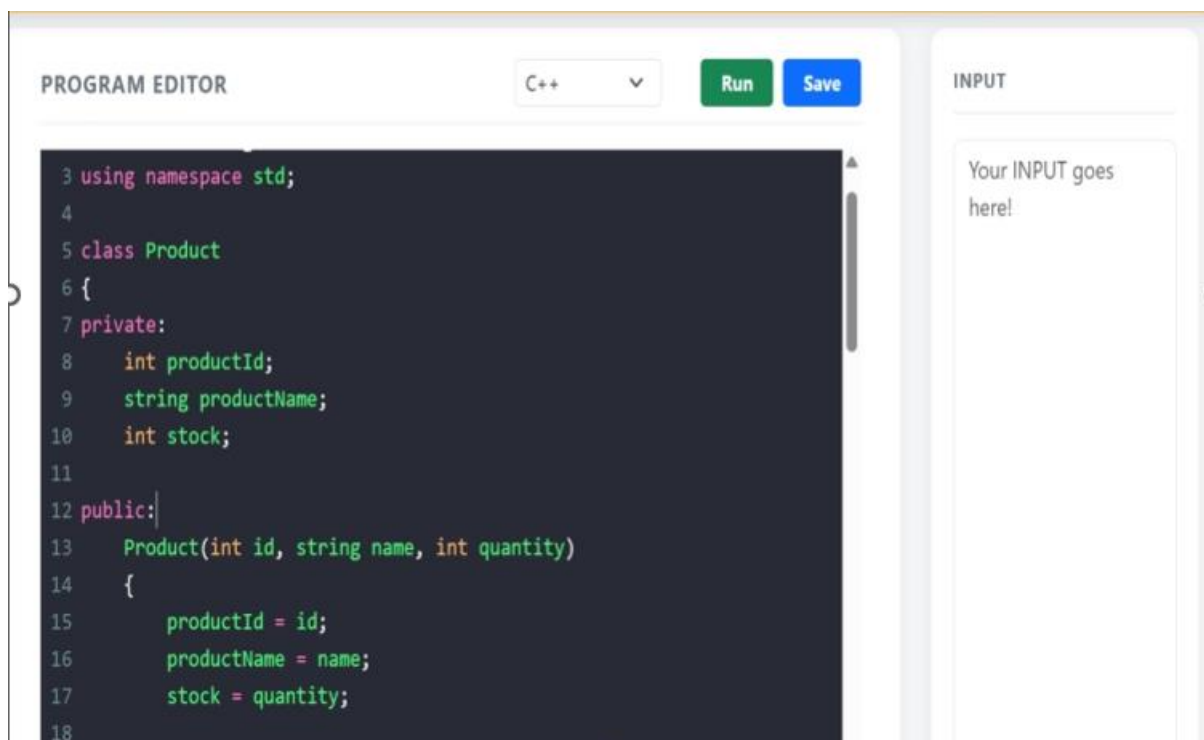
```

```

Product p1(101, "Laptop", 20);
Product p2(102, "Mouse", 30);
cout << "\n--- Product 1 ---";
p1.display();
cout << "\n--- Product 2 ---";
p2.display();
cout << "\n--- Copying Product 1 ---";
Product p3(p1);
cout << "\n--- Copied Product ---";
p3.display();
cout << "\n--- Adding Products ---";
Product p4 = p1 + p2;

cout << "\n--- Combined Product ---";
p4.display();
cout << "\n===== END OF PROGRAM =====\n"
return 0;

```



OUTPUT

```
===== INVENTORY PRODUCT MANAGEMENT SYSTEM =====

Parameterized Constructor called for Laptop
Parameterized Constructor called for Mouse

--- Product 1 ---
Product ID    : 101
Product Name  : Laptop
Stock        : 20
```

```
3)#include <iostream>
#include <string>
using namespace std;
class Product
{
private:
    int id;
    string name;
    int quantity;
public:
    Product(int i, string n, int q)
    {
        id = i;
        name = n;
        quantity = q;
        cout << "Product created: " << name << endl;
    }
    Product(const Product &p)
    {
        id = p.id;
```

```

        name = p.name;
        quantity = p.quantity;
        cout << "Product copied: " << name << endl;
    }
    Product operator+(const Product &p)
    {
        return Product(
            id,
            name + " + " + p.name,
            quantity + p.quantity
        );
    }
    void display()
    {
        cout << "\nProduct ID: " << id << endl;
        cout << "Product Name: " << name << endl;
        cout << "Quantity: " << quantity << endl;
    }
    ~Product()
    {
        cout << "Destructor called for: " << name << endl;
    }
};

```

```

int main()
{
    int id1, id2;
    int qty1, qty2;
    string name1, name2;

```

```
cout << "===== SHOPPING CART MANAGEMENT SYSTEM =====" << endl;
cout << "\nEnter Product 1 ID: ";
cin >> id1;

cout << "Enter Product 1 Name: ";
cin >> name1;
cout << "Enter Product 1 Quantity: ";
cin >> qty1;
cout << "\nEnter Product 2 ID: ";
cin >> id2;
cout << "Enter Product 2 Name: ";
cin >> name2;
cout << "Enter Product 2 Quantity: ";
cin >> qty2;
Product p1(id1, name1, qty1);
Product p2(id2, name2, qty2);
cout << "\n===== PRODUCT DETAILS =====";
p1.display();
p2.display();
cout << "\n===== COPY CONSTRUCTOR =====";
Product p3(p1);
p3.display();
cout << "\n===== MERGING PRODUCTS =====";
Product p4 = p1 + p2;
p4.display();
cout << "\n===== PROGRAM END =====" << endl;
return 0;
```

PROGRAM EDITOR

C++



Run

Save

```
3 using namespace std;
4
5 class Product
6 {
7 private:
8     int productId;
9     string productName;
10    int stock;
11
12 public:
13    Product(int id, string name, int quantity)
14    {
15        productId = id;
16        productName = name;
17        stock = quantity;
18    }
```

INPUT

Your INPUT goes
here!

OUTPUT

```
Quantity: Enter Product 2 ID: Enter Product 2 Name: Enter Product
2 Quantity: Constructor called: TV
Constructor called:

Product 1:

Product Details
ID       : 1011112222
Name     : TV
Quantity : 4
```